



Alice Giubergia

PHD IN INFORMATION ENGINEERING | Neuroimaging & Data Science

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PROFILE

PhD-level neuroimaging engineer with 5+ years of experience turning large, noisy 4D imaging datasets into validated, reproducible analysis pipelines. Doctoral research centered on fMRI acquisition and processing methodology, benchmarking multi-echo vs single-echo EPI, denoising strategies and functional-connectivity estimation, published as first author in three peer-reviewed journals. Comfortable owning a problem end to end: experimental design, data acquisition in clinical settings, Python/MATLAB pipeline development, statistical modelling, and communicating results to clinical and non-technical stakeholders.

TECHNICAL SKILLS

Programming: Python (NumPy, pandas, nibabel, nilearn, scikit-learn), MATLAB, bash / Linux shell scripting, Git & GitHub, SQL (basic), C++ (basic), LaTeX.

Neuroimaging tools: SPM, FSL, FreeSurfer, ANTs, LayNii, Segmentator, ART (Artifact Detection Tools), Presentation®, fMRIPrep, BIDS, Docker.

Methods: preprocessing and denoising (ICA-based nuisance regression, motion-parameter PCA, bandpass filtering), multi-echo EPI, GLM and task-evoked analysis, functional connectivity, laminar / layer-specific fMRI at ultra-high field (7T), atlas- and peak-based ROI definition, group-level statistical inference.

Machine learning & statistics: machine-learning analysis of multi-domain clinical data (imaging, genetic, behavioral); artificial neural networks for real-time signal classification; statistical modelling and group-level inference.

Clinical & data practice: fMRI protocol set-up and acquisition in pediatric and neonatal settings, quality control on clinical data, interdisciplinary work with clinicians and psychologists, GDPR-compliant handling of clinical.

Languages: Italian (native), English (fluent, peer-reviewed publications and international conferences).

Curriculum Vitae

RESEARCH EXPERIENCE

Research Engineer — Neuroimaging Lab | Mar 2026 – Present

IRCCS "Eugenio Medea" — Associazione La Nostra Famiglia | Bosisio Parini (LC), Italy

- Built an end-to-end laminar (layer-specific) fMRI pipeline for ultra-high-field data using ANTs, FreeSurfer and LayNii, producing group-level cortical-depth activation profiles across ROIs.
- Designed a subject-fingerprinting analysis on a fMRI+EEG sleep dataset: state-transition matrices from EEG-scored 30-second epochs, cost-based distance metrics, and classification validation on held-out runs.

PhD in Information Engineering – Curriculum Bioengineering | Nov 2022 – Mar 2026

Università degli Studi di Padova | research carried out at IRCCS "Eugenio Medea", Lecco

- Benchmarked multi-echo against single-echo EPI acquisition for task fMRI and quantified how preprocessing choices propagate into functional-connectivity estimates; both studies published as first author (Imaging Neuroscience; Journal of Neuroscience Research).
- Designed, implemented and validated an fMRI paradigm for visually degraded word reading, from task programming to MR data collection; results published as first author in Behavioral and Brain Functions.
- Visiting researcher, IRCCS Stella Maris from March 2025 to June 2025. Ran a pilot ultra-high-field (7T) laminar fMRI study on reading, supported by a competitive AIRMM travel grant.

Assistant Research Engineer — Neuroimaging Lab | Jun 2021 – Sep 2022

IRCCS "Eugenio Medea" — Associazione La Nostra Famiglia | Bosisio Parini (LC), Italy

- Integrated analysis of multi-domain data (imaging, genetic, behavioural) with machine-learning methods in a pilot project on developmental dyslexia.

PhD in Information Engineering (Bioengineering) | Nov 2022 – Mar 2026

Università degli Studi di Padova, Padova, Italy

Thesis: "Functional MRI data processing and analysis: novel approaches with application to atypical reading" — Supervisor: Alessandra Bertoldo

EDUCATION

MSc in Biomedical Engineering | Sep 2018 – Dec 2020

Politecnico di Milano, Milano, Italy

Thesis: real-time ANN-based gesture recognition (Microsoft Kinect skeleton data) for humanoid-robot therapy with children with ASD, in collaboration with CARELab, Fondazione Don Gnocchi — Supervisor: Alessandra Pedrocchi

BSc in Biomedical Engineering | Sep 2015 – Sep 2018

Politecnico di Torino, Torino, Italy

Thesis: analysis of competing mammography / DBT CAD and breast-density assessment systems (internship at IM3D Clinic Piemonte)

PERSONAL SKILLS

- Set up and acquisition of an fMRI experiment, from task implementation to MR data gathering, in a pediatric setting (IRCCS "Eugenio Medea" (LC))
- Technical support to clinical trials with patients, skill acquired during my master thesis period at CARELab (Fondazione Don Gnocchi – Milano)
- Data analysis for clinical trials, skill acquired during my master thesis project
- Ability to work in team, skill I acquired developing my master thesis project with a colleague
- Ability to work in an interdisciplinary team, especially with clinicians, skill acquired during my thesis work and my experience at IRCCS "Eugenio Medea" (LC)

PUBLICATIONS

Selected:

- **Giubergia, A.**, Mascheretti, S., Lampis, V., et al. (2026). An examination of task-evoked functional MRI data processing in functional connectivity. *Journal of Neuroscience Research*.
- **Giubergia, A.**, Lampis, V., Mauri, C., et al. (2026). Neural compensation through dorsal stream involvement in visual word recognition under perceptual degradation in a transparent language: evidence from fMRI. *Behavioral and Brain Functions*.
- **Giubergia, A.**, Ferrazzi, G., Castellaro, M., et al. (2025). Multi-echo versus single-echo EPI sequences for task-fMRI: a comparative study. *Imaging Neuroscience*, 3.
- Peruzzo, D., Ciceri, T., Mascheretti, S., et al. incl. **Giubergia, A.** (2024). Brain alteration patterns in children with Duchenne muscular dystrophy: a machine-learning approach to MRI. *Journal of Neuromuscular Diseases*.
- Ivani, A. S., **Giubergia, A.**, Santos, L., et al. (2022). A gesture recognition algorithm in a robot therapy for ASD children. *Biomedical Signal Processing and Control*, 74, 103512.

AWARDS

- Best oral presentation, XV AIRMM National Congress, Padova, April 2024.
- AIRMM Travel Grant (Associazione Italiana di Risonanza Magnetica in Medicina) for the project "Neural mechanisms of reading: cortical layer interactions and plasticity — insights from ultra-high-field MRI".

PERSONAL STATEMENT

I hold a Ph.D. in Information Engineering (Curriculum Bioengineering) from the University of Padova, which I completed in 2026, and an M.Sc. in Biomedical Engineering from Politecnico di Milano. My doctoral research focused on the development and evaluation of novel methodological approaches in functional MRI, spanning acquisition protocols, preprocessing pipelines, and functional connectivity estimation, with application to the neuroscience of atypical reading.

My interest in neuroimaging research grew during my years as a Research Assistant Engineer at the Neuroimaging Lab of IRCCS "Eugenio Medea" (Lecco), where I built a solid foundation in fMRI signal processing, statistical analysis, and the handling of clinical data. A visiting period at IRCCS Stella Maris further enriched my experience and broadened my perspective on collaborative, clinically oriented research.

Throughout my career I have developed strong skills in fMRI methodology, from multi-echo acquisition and advanced denoising strategies to functional connectivity modelling and ultra-high-field imaging, alongside a genuine commitment to rigorous, reproducible science. I am a consistent problem-solver, a fast learner, and an effective collaborator, always looking to contribute to and grow within a stimulating research environment.

I am currently seeking new opportunities to deepen my expertise in neuroimaging and to continue contributing to the advancement of fMRI methods and their application to cognitive and clinical neuroscience.

FULL PUBLICATIONS

International Journals

- Giubergia, A., Mascheretti, S., Lampis, V., Ciceri, T., Villa, M., Andreola, C., Arrigoni, F., Bertoldo, A. & Peruzzo, D. (2026). An examination of task-evoked functional MRI data processing in functional connectivity. *Journal of Neuroscience Research*. <https://doi.org/10.1002/jnr.70132>
- Giubergia, A., Lampis, V., Mauri, C., Montano, F., Ferrazzi, G., Castellaro, M., Bertoldo, A., & Peruzzo, D. (2026). Neural compensation through dorsal stream involvement in visual word recognition under perceptual degradation in a transparent language: evidence from fMRI. *Behavioral and Brain Functions : BBF*, 10.1186/s12993-026-00330-4. Advance online publication. <https://doi.org/10.1186/s12993-026-00330-4>.
- Mascheroni, E., Peruzzo, D., Giubergia, A., Cordolcini, L., Ciceri, T., Mariani Wigley, I.L., Castagna, A., Riva, B., Montano, F., Agarwal, N., & Montiroso, R. (2025). Brain response to CT-related touch in infants: Contribution of temperament and maternal touch. *Journal of experimental child psychology*, 263, 106419 .
- Giubergia, A., Ferrazzi, G., Castellaro, M., Mascheretti, S., Lampis, V., Montano, F., Bertoldo, A., & Peruzzo, D. (2025). Multi-echo versus single-echo EPI sequences for task-fMRI: A comparative study. *Imaging Neuroscience*, 3.
- Maccioni, L., Giacomel, A., Pinamonti, M., De Francisci, M., Del Pup, F., Giubergia, A., Vallini, G., Visani, V., Barzon, L., Dipasquale, O., Bertoldo, A., Moretto, M., & Veronese, M. (2025). Quantitative neuroimaging meets normative modelling: The last mile for precision medicine applications (Preprint v1). *Preprints.org*. <https://doi.org/10.20944/preprints202509.2335.v1>
- Mascheretti, S., Arrigoni, F., Toraldo, A., Giubergia, A., Andreola, C., Villa, M., Lampis, V., Giorda, R., Villa, M., & Peruzzo, D. (2024). Alterations in neural activation in the ventral frontoparietal network during complex magnocellular stimuli in developmental dyslexia associated with READ1 deletion. *Behavioral and Brain Functions: BBF*, 20.
- Peruzzo, D., Ciceri, T., Mascheretti, S., Lampis, V., Arrigoni, F., Agarwal, N., Giubergia, A., Villa, F.M., Crippa, A., Nobile, M., Mani, E., Russo, A., & D'Angelo, M. (2024). Brain Alteration Patterns in Children with Duchenne Muscular Dystrophy: A Machine Learning Approach to Magnetic Resonance Imaging. *Journal of neuromuscular diseases*.
- Ciceri, T., Squarcina, L., Giubergia, A., Bertoldo, A., Brambilla, P., & Peruzzo, D. (2023). Review on deep learning fetal brain segmentation from Magnetic Resonance images. *Artificial intelligence in medicine*, 143, 102608.
- Santos, L., Annunziata, S., Geminiani, A., Ivani, A.S., Giubergia, A., Garofalo, D., Caglio, A., Brazzoli, E., Lipari, R., Carrozza, M.C., Ambrosini, E., Olivieri, I., & Pedrocchi, A.L. (2023). Applications of Robotics for Autism Spectrum Disorder: a Scoping Review. *Review Journal of Autism and Developmental Disorders*.
- Ivani, A.S., Giubergia, A., Santos, L., Geminiani, A., Annunziata, S., Caglio, A., Olivieri, I., & Pedrocchi, A.L. (2022). A gesture recognition algorithm in a robot therapy for ASD children. *Biomed. Signal Process. Control.*, 74, 103512.

Curriculum Vitae

International Conferences

A. Giubergia, G. Ferrazzi, M. Castellaro, S. Mascheretti, V. Lampis, F. Montano, A. Bertoldo, D. Peruzzo, "Comparison of Multi-Echo and Single-Echo EPI sequences for task-fMRI", 40th annual scientific meeting ESMRMB, October 2024, Barcelona, Spain

A. Giubergia, S. Mascheretti, V. Lampis, T. Ciceri, M. Villa, C. Andreola, F. Arrigoni, A. Bertoldo, D. Peruzzo, "How to shape connectivity in task-evoked Functional MRI data: An examination.", OHBM 2024 Annual Meeting, June 2024, Seoul, Korea

A. Giubergia, S. Mascheretti, F. Arrigoni, A. Toraldo, C. Andreola, M. Villa, V. Lampis, R. Giorda, M. Villa, D. Peruzzo, "Genotype-Phenotype interaction in the attentive network in developmental dyslexia a population of children with Developmental Dyslexia.", OHBM 2024 Annual Meeting, June 2024, Seoul, Korea

A. Giubergia, S. Mascheretti, F. Arrigoni, A. Toraldo, C. Andreola, M. Villa, V. Lampis, R. Giorda, M. Villa, D. Peruzzo, "Aberrant activations associated with developmental dyslexia and READ1 deletion induced by visual/attentive tasks.", ISMRM 33rd Annual Meeting & Exhibition, May 2024, Singapore

F. Maccarone, M. Bonino, A. Giubergia, S. Mascheretti, V. Lampis, C. Mauri, N. Agarwal, D. Peruzzo, "An fMRI single subject's study with reading impairment – a new paradigm towards personalised medicine.", ISMRM 33rd Annual Meeting & Exhibition, May 2024, Singapore

D. Peruzzo, A. Giubergia, Tommaso Ciceri, Francesca Maccarone, Isabella Mariani Wigley, Annalisa Castagna, Rosario Montiroso, Eleonora Mascheroni, 2023, "Neural activations induced through affective touch in newborn infants", 2023 OHBM Annual Meeting, Montreal, Canada, 22-26 July 2023

A. Giubergia, S. Mascheretti, V. Lampis, T. Ciceri, F. Maccarone, M. Villa, C. Andreola, F. Arrigoni, A. Bertoldo, D. Peruzzo, "Pseudo resting-state by task-evoked functional connectivity.", ISMRM 32nd Annual Meeting & Exhibition, June 2023, Toronto, Canada

National Conferences

A. Giubergia, S. Mascheretti, F. Arrigoni, A. Toraldo, C. Andreola, M. Villa, V. Lampis, R. Giorda, M. Villa, D. Peruzzo, "Peculiar activations linked to developmental dyslexia and READ1 deletion prompted by visual or attentive tasks.", XV Congresso Nazionale AIRMM, April 2024, Padova, Italy

A. Giubergia, S. Mascheretti, V. Lampis, T. Ciceri, F. Maccarone, M. Villa, C. Andreola, F. Arrigoni, A. Bertoldo and D. Peruzzo, "Pseudo resting-state investigation from task-evoked functional MRI signals", VIII National Congress of Bioengineering, June 2023, Padova, Italy

Ai sensi del GDPR e del Regolamento Europeo 679/2016, autorizzo al trattamento dei dati personali contenuti all'interno di questo curriculum vitae

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